

20 Important Array Programs - Problem Statements with Input/Output

1. Find the largest element in an array

Input: {10, 25, 3, 99, 45}

Output: Largest element = 99

2. Find the smallest element in an array

Input: {10, 25, 3, 99, 45}

Output: Smallest element = 3

3. Find the second largest and second smallest elements

Input: {10, 25, 3, 99, 45}

Output: Second Largest = 45, Second Smallest = 10

4. Sort an array (ascending & descending order)

Input: {12, 5, 8, 1, 9}

Output:

Ascending: {1, 5, 8, 9, 12}

Descending: {12, 9, 8, 5, 1}

5. Reverse an array

Input: {10, 20, 30, 40, 50}

Output: {50, 40, 30, 20, 10}

6. Find the sum and average of array elements

Input: {5, 10, 15, 20}

Output: Sum = 50, Average = 12.5

7. Copy elements of one array into another

Input: {1, 2, 3, 4, 5}

Output: Copied array = {1, 2, 3, 4, 5}

8. Search an element in an array (Linear Search)

Input: Array = {10, 20, 30, 40}, Element = 30

Output: Element 30 found at index 2

9. Search an element in a sorted array (Binary Search)

Input: Array = {5, 10, 15, 20, 25}, Element = 15

Output: Element 15 found at index 2

10. Merge two arrays into a single array

Input: {1, 2, 3} and {4, 5, 6}
Output: {1, 2, 3, 4, 5, 6}

11. Remove duplicate elements from an array

Input: {1, 2, 2, 3, 4, 4, 5}
Output: {1, 2, 3, 4, 5}

12. Count the frequency of each element in an array

Input: {10, 20, 10, 30, 20, 10}
Output:
10 → 3 times
20 → 2 times
30 → 1 time

13. Find missing number in a sequence (1 to n)

Input: {1, 2, 3, 5}
Output: Missing number = 4

14. Find duplicate elements in an array

Input: {2, 5, 7, 2, 9, 5}
Output: Duplicate elements = {2, 5}

15. Find the occurrence of a particular element

Input: Array = {1, 2, 3, 2, 4, 2}, Element = 2
Output: Element 2 occurs 3 times

16. Rotate an array left/right by 'k' positions

Input: {1, 2, 3, 4, 5}, k = 2
Output:
Left Rotation: {3, 4, 5, 1, 2}
Right Rotation: {4, 5, 1, 2, 3}

17. Check if two arrays are equal

Input: {1, 2, 3, 4} and {1, 2, 3, 4}
Output: Arrays are equal

18. Find the intersection of two arrays

Input: {1, 2, 3, 4} and {3, 4, 5, 6}
Output: Intersection = {3, 4}

19. Find the union of two arrays

Input: {1, 2, 3} and {3, 4, 5}
Output: Union = {1, 2, 3, 4, 5}

20. Find the pair of elements whose sum is equal to a given number (Two-Sum problem)

Input: Array = {2, 7, 11, 15}, Target = 9
Output: Pair = (2, 7)